

Source Mixing and Separation Robust Audio Stegenography¹²

Speaker: Gewenxin Yu

Queen Mary University of London

March 2023

¹Takahashi, N., Singh, M.K., & Mitsufuji, Y. (2021). [ArXiv Link](#).

²This is the presentation for module ECS7013P. The link to the companion video is [here](#).

The First Two Books

Hypnerotomachia Poliphili = *The strife of love in a dream*,
Anonymous, 1499

Steganographia, Trithemius (1462-1516), 1499

³History of steganography and cryptography, The information hiding homepage: [Link](#).

The Book Hypnerotomachia Poliphili

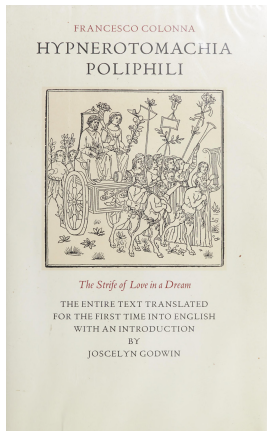


Figure: Hypnerotomachia Poliphili, Anonymous, 1499. ⁴

⁴Extract From Archive.org. [The Link](#).

What is Audio Steganography?⁵

The science of concealing secret messages inside a host audio called a *carrier* in such a way that the concealment is **unnoticeable to human ears**.

⁵D. C. Kar and C. J. Mulkey, A multi-threshold based audio steganography scheme, Journal of Information Security and Applications, vol. 23, pp. 5467, 2015.

- Kreuk et al. *Hide&Speak* ⁶
 - The method demonstrates robustness against some types of distortion that can be introduced during data transmission.
 - Does not assume that the carrier could be aggressively edited.
 - Issue: it is difficult to conceal messages in each source in a mixture.
 - (This paper will address this issue.)

⁶F. Kreuk, Y. Adi, B. Raj, R. Singh, and J. Keshet, Hide and speak: Towards deep neural networks for speech steganography, in Proc. Interspeech, 2020.

Proposed Method

Components

- Concealer: $E(c, m) = \hat{c}$
- Decoder: $D(\hat{c}) = \hat{m}$
- Goal: Minimize the message reconstruction error $d_m(m, \hat{m})$

Loss Function

$$L(c, m) = \lambda_c d_c(c, \hat{c}) + \lambda_m d_m(m, \hat{m}), \quad (1)$$

Here, λ_c and λ_m are weights, and $d_c(c, \hat{c})$ is a metric used to measure the similarity between the original and embedded carriers. The l_1 norm is used for both d_c and d_m .

MSRAS Model Overview

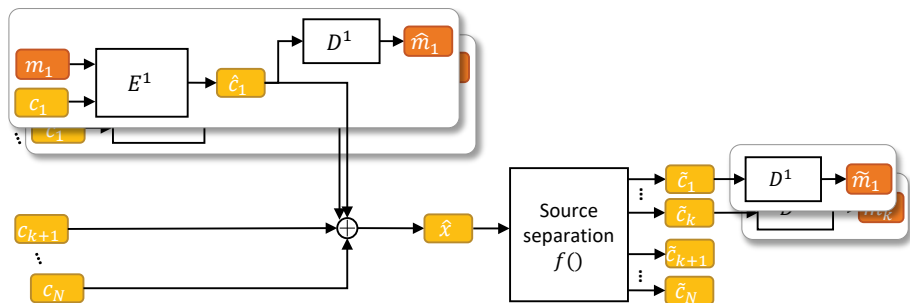


Figure: Model overview.

$$L^{(i)} = \lambda_1^i d_c(c_i, \hat{c}_i) + \lambda_2^i d_m(m_i, \hat{m}_i) + \lambda_3^i d_m(m_i, \tilde{m}_i), \quad (2)$$

where $\tilde{m}_i = D^i(f(\hat{x})_i)$ denotes the message recovered from the i th separated source $f(\hat{x})_i$.

- The reconstructed message $\tilde{m}$ depends on all sources in the mixture $\hat{x}$, while the loss $L^{(i)}$ is also affected by other concealers $E^{j \neq i}$.
- Using the summation of all losses $L^{all} = \sum_{i \in \Gamma} L^{(i)}$ universally may not be optimal.

Algorithm 1: Multi-source concealer decoder learning

```
1: for (# iteration) do
2:   sample  $c_1, \dots, c_N$  and  $m_1, \dots, m_k$ 
3:   for  $i = 1, \dots, k$  (source types) do
4:     // conceal messages and mix
5:      $\hat{x} = \sum_{j=1}^k E^j(c, m_j) + \sum_{j=k+1}^N c_j$ 
6:     // decode messages from  $i$ th embedded source and separation
7:      $\hat{m}_i = D^i(\hat{c}), \tilde{m}_i = D^i(f(\hat{x})_i)$ 
8:     freeze model parameters of  $E^{j \neq i}, D^{j \neq i}$  and update the parameters of  $E^i$  and  $D^i$  using  $L^{(i)}$  in (2)
9:   end for
10: end for
```

Concealing Message In Time Domain: TCD Model

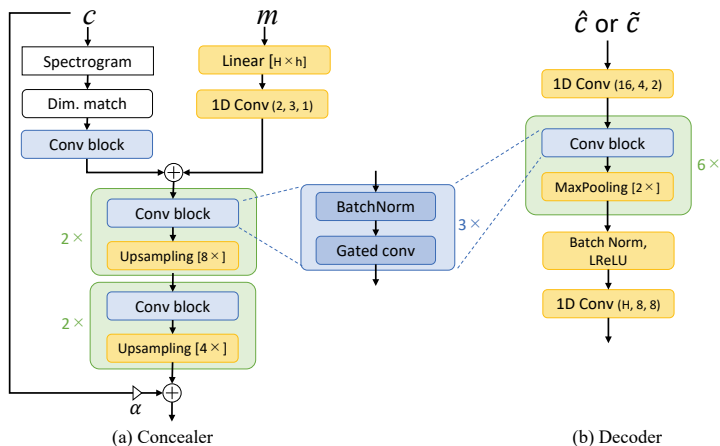


Figure: TCD model architecture. Conv (o, κ, s) denotes the convolution with o filters, kernel size κ , and stride s .

- Learning the TCD through the source separation model is extremely challenging.
- The models are unable to learn the steganographic function from scratch by directly using the previous Loss Function.

1. Train the concealer to generate audio similar to the carrier using the multi-resolution downsampled spectral (MDS) loss.
2. Recover the message from the modified carrier.
3. Incorporate source separation into the training criteria.
4. Gradually increasing α to 1.
5. Using multi-resolution downsampled spectrogram ratio (MDSR) for $d_c()$.

Step	λ_1	λ_2	λ_3	α	d_c	# iteration
1	1	0	0	0	MDS	500
2	0.1	1	0	0	MDS	2500
3	0.1	1	1	0	MDS	5000
4	0.1	1	1	0 $\rightarrow$ 1	MDS	2000
5	0.0002	1	1	1	MDSR	5000

Table: Curriculum for model training.

Experiments

Experimental Setup

- Dataset: MUSDB18⁷ (100 songs for training, 50 for testing)
- Each song has four sources (*bass, drums, other, vocals*) recorded in stereo format at 44.1 kHz.
- Carrier: *Drum* track
- Source separation model: Demucs⁸
- Loss function: Cross-entropy loss to minimize message reconstruction error ($d_m(m, \hat{m})$)
- Optimizer: Adam optimizer with learning rate of 0.001

⁷A. Liutkus, F.-R. Stöter, and N. Ito, The 2018 signal separation evaluation campaign, in Proc LVA/ICA, 2018.

⁸A. Deffossez, N. Usunier, L. Bottou, and F. Bach, Music source separation in the waveform domain, arXiv preprint arXiv:1911.13254, 2019.

1. The state-of-the-art method from Kreuk et al. *Hide&Speak*.
2. The loss function $L^{(i)}$ (1) .
3. The TCD Model is trained using the conventional loss function $L(c, m)$ (2).
4. The proposed model is trained only on the separated source.

Results: SNR and accuracy

Model	SNR [dB]		Accuracy [%]	
	$\hat{c}$	$\tilde{c}$	$D(\hat{c})$	$D(\tilde{c})$
Hide&Speak	30.5	25.9	100.0	36.0
TCD trained with (1)	35.3	40.3	100.0	3.6
Hide&Speak + $d_m(m_i, \tilde{m}_i)$	31.6	25.3	99.6	74.7
MSRAS ($\lambda_2 = 0$)	35.3	37.8	61.5	100.0
MSRAS (Proposed)	35.9	34.5	100.0	100.0

Table: SNR and accuracy of different methods.

Results: Subjective detectability

Evaluated source	Accuracy [%]
Embedded source ($\hat{c}$)	55.5
Separated source ($\tilde{c}$)	53.9

Table: *Subjective test on perceptibility of the perturbations.*

Results: Omitting steps in curriculum learning

Step	SNR [dB]		Accuracy [%]	
	$\hat{c}$	$\tilde{c}$	$D(\hat{c})$	$D(\tilde{c})$
2,3,4,5	54.1	52.1	6.3	6.3
1,3,4,5	74.3	70.9	6.3	6.3
1,2,4,5	27.2	26.3	100.0	100.0
1,2,3,5	27.4	28.2	100.0	100.0
1,2,3,4	18.3	20.9	100.0	100.0
All	35.9	34.5	100.0	100.0

Table: Comparison of different curricula.

Results: Interference from multiple models

Instruments	SNR [dB]		Accuracy [%]	
	$\hat{c}$	$\tilde{c}$	$D(\hat{c})$	$D(\tilde{c})$
drums	27.9	30.8	100.0	100.0
vocals	32.3	29.9	86.8	85.6

Table: *Multiple models.*

Results: Robustness against noise and edits

Distortion	w/o D.A.	w/ D.A.
AGN ($\sigma = 0.001$)	99.9	97.2
Ch. drop	3.4	97.0
EQ	100.0	97.1
MP3 compression 320k	3.4	95.0
MP3 compression 128k	3.7	94.9

Table: *Robustness of models trained with and without data augmentation evaluated with different types of distortion.*

- Proposed MSRAS, a robust audio steganography technique based on source mixing and separation.
- Developed a DNN-based concealer that operates in the time domain to avoid distortions.
- Achieved high performance in message recovery from both unprocessed and separated sources.
- Enabled independent message concealment without interference with other sources.

Limitations

- Comparison with more existing methods
- Security analysis
- Real-world applicability

Thank You!